$$\begin{aligned} q' \in I_{[E]}(v) &\rightarrow q' \in I_{[E]}(q.w'_q.\widehat{e_q}) && \text{by Lemma 4.2(1) and definition of } notdep \\ &\xrightarrow{*} q' \in I_{[E]}(q.w') && \text{by Lemma 5.1.2a} \qquad (15) \\ &\rightarrow q' \in I_{[E]}(q.w_R.v') && \text{by (14), Lemma 5.1.1a} \\ &\rightarrow q' \in WI_{[E]}(q.w_R) && \text{by Lemma 4.2(2)} \end{aligned}$$

The algorithm then guarantees (in lines 8 and 9) that there is one such $q' \in I_{[E]}(v)$ which is added to *backtrack*(E). Additionally, by (15) we have that no other event happens-before q' in $E.q.w'$. Therefore $q' \notin$ *init_sleep*(E), otherwise it would have stayed in the sleep set (transferred by line 10) and could not have been chosen for execution.

Together, $q' \in$ *backtrack*(E) and $q' \notin$ *init_sleep*(E) imply that $q' \in$ *done*(E) but this contradicts Claim 5.5. Such a sequence w can therefore not exist and this concludes the proof of Claim 5.4. □

Moving on with the proof of Lemma 5.3. By Claim 5.4, for any $E.w$, we have $p \in WI_{[E]}(w)$ for some $p \in$ *done*(E). Let q be the process among these which is explored first from $s_{[E]}$. We intend to apply the inductive hypothesis for $E.q$ and $w \setminus q$:

IH-condition 5.7. $(w \setminus q) \in$ *to_be_explored*$(E.q)$.

Proof of IH-condition 5.7. We must establish two properties:(i) that $E.q \vdash w \setminus q$ and (ii) that $I_{[E.q]}(w \setminus q) \cap$ *init_sleep*$(E.q) = \emptyset$. We infer $E.q \vdash w \setminus q$ from $E \vdash w$ and $q \in WI_{[E]}(w)$. For the second property, $I_{[E.q]}(w \setminus q)$ may extend $I_{[E]}(w)$ only by processes $q' \in w$ such that $next_{[E]}(q) \rightarrow_{E.w} \langle q', j\rangle$, where $\langle q', j\rangle$ is the first event of q' in w. However, any such process q' cannot be in *init_sleep*$(E.q)$, since in this case $E \not\vdash q \diamond q'$ so q' is removed from the sleep set when exploring $E.q$. Given also that *init_sleep*$(E.q) \setminus$ *init_sleep*(E) does not contain any process in $I_{[E]}(w)$ (recall that q was the first process in $I_{[E]}(w)$ to be explored), we infer IH-condition 5.7. □

The conclusion from the inductive hypothesis is that the algorithm explores some sequence $E.q.\hat{w}$ which is in $[E.q.(w \setminus q).v]_{\simeq}$ for some v. We have two cases:

- $q \in w$. From $q \in WI_{[E]}(w)$ we infer $E.q.(w \setminus q).v \simeq E.w.v$, which establishes the lemma.
- $q \notin w$, i.e., $(w \setminus q) = w$. From $E \vdash q \diamond w$, we infer $E.q.\hat{w} \simeq E.q.w.v \simeq E.w.q.v$.

This concludes the proof of Lemma 5.3. □

Using Lemma 5.3 we can finally conclude the proof of the main theorem:

Proof of Theorem 5.2. Since Algorithm 1 is initially called with *Explore*$(\langle\rangle, \emptyset)$, Lemma 5.3 implies correctness in the sense that for all execution sequences E the algorithm explores some execution sequence E' which is in $[E.v]_{\simeq}$ for some v. In particular, if E is maximal, then $E' \simeq E$. □

6 WAKEUP TREES

As we described earlier, *source-DPOR* may still lead to sleep-set blocked explorations. We therefore present an algorithm, called *optimal-DPOR*, which is provably optimal in that it always explores exactly one interleaving per Mazurkiewicz trace, and never encounters sleep-set blocking. Optimal-DPOR is obtained by combining source sets with a novel mechanism, called *wakeup trees*, which control the initial steps of future explorations.

Wakeup trees can be motivated by looking at lines 6–9 of Algorithm 1. At these lines, it is found that some execution sequence starting with $E'.v$ should be performed in order to reverse a detected race. However, at line 9, only a single process from the sequence v is entered into *backtrack*(E'), thus "forgetting" information about how to reverse this race. Since the new exploration after $E'.q$